

Table probe

I. BODY TABLES

Each table below is flushed onto its own page, so the log attributes every overfull box to the file that was open when it was set.

TABLE I: Hands that can be obtained: sold, open-source, or with no release status stated. 20 rows, sorted by joint count; a row whose joint count no source states sorts last rather than being ranked by a number nobody stated. 68 of the 160 cells in the 8 specification columns (DoF through price) (42%) are —: no source stated the value. The key, maker, open-hardware, status and usage columns are excluded from that count because they are never blank, and counting them would dilute the finding. *actuation*, *force kind* and *tactile* are controlled vocabularies; the source sentence each was read from is in Table XI. *force N* is not a ranking: read *force kind* first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor number are different measurements. *uses* counts the method rows whose own experiments run on that hand. Every figure is the maker’s or the authors’ own claim.

hand key	maker	DoF	act. DoF	actuation	mass g	force N	force kind	tactile	price USD	open HW	status	uses
shadow_dexterous_hand_2005	Shadow Robot Co.	24	20	tendon	4 300	—	—	optional	—	no	sold	21
proception_prohand_2026	Proception Inc (YC W25)	22	—	tendon	—	—	—	unstated	—	no	sold	0
sharpa_wave_2026	Sharpa Robotics (Sharpa Pte Ltd)	22	22	—	1 300	20 / 12	grip	unstated	—	no	sold	6
bidexhand_2025	Northwestern Univ.	21	16	tendon	—	2.14	fingertip normal	none	—	yes	open-source	0
ruka_v2_2026	NYU and NYU Shanghai	20	16	tendon	—	—	—	optional	1 500	yes	open-source	0
tesollo_dg5f_2024	Tesollo Inc. (Incheon HQ,...)	20	20	direct drive	1 763	—	—	none	—	no	sold	0
unitree_dex5_2025	Unitree Robotics (Yushu Tech....)	20	16	tendon	1 100	10	fingertip normal	array	—	no	sold	0
wuji_hand_2025	Wuji Tech. (founded 2019)	20	20	direct drive	—	—	—	fingertip	—	no	sold	2
orca_hand_2025	ETH Zurich	17	17	tendon	1 200	—	—	fingertip	—	yes	open-source	0
allegro_hand_v4_2016	Wonik Robotics Co. Ltd. (Seoul,...)	16	16	—	—	—	—	none	—	no	sold	35
leap_hand_2023	CMU	16	16	direct drive	595	19.5	pull-out	none	2 000	yes	open-source	11
ruka_2025	NYU	15	11	tendon	—	2.74	pinch	none	1 300	yes	open-source	0
inspire_rh56dfx_2023	Beijing Inspire Robots Tech. Co.,...	12	6	—	540	10	vendor, no protocol	none	—	no	sold	19
robotera_xhandl_2024	ROBOTERA	12	12	geared	1 100	—	—	unstated	14 000	no	sold	8
shadow_dex_ee_2024	Shadow Robot Co., in...	12	—	—	4 100	—	—	unstated	—	no	sold	1
brainco_revo2_2025	BrainCo Inc.	11	6	—	383	15	pinch	unstated	—	no	sold	1
agibot_omnihand_2025	—	—	—	—	—	—	—	—	—	—	unstated	1
dexhand_open_source_2023	IoT Design Shop	—	—	tendon	—	—	—	none	300	yes	open-source	0
linkerbot_120_2025	LinkerBot (SDK author 'CHIUS INC') USA)	—	—	—	—	—	—	unstated	—	no	sold	1
psyonic_ability_hand_2021	—	—	6	linkage	—	—	—	unstated	—	no	sold	7

TABLE II: Hands announced, prototyped or held internally, which cannot be bought or built. 13 rows, the same columns as Table I plus the date of the claim and the class of artefact the claim was read off: *datasheet* for a paper, datasheet or vendor specification page, *video* for a demonstration reel, *press* for a press release or a third-party tracker. 60 of the 104 cells in the 8 specification columns (DoF through price) (58%) are –: no source stated the value. That share is far higher than in Table I and it is the finding of this table rather than a defect of it: an unreleased hand is announced without a specification. *claim date* prints *undated* where the source page carries no date of its own; a fetch date is not a claim date.

hand key	maker	DoF	act. actuation DoF	mass g	force N	force kind	tactile	price USD	open HW	status	uses	claim date	evidence
clone_ robotics_ hand_2024	Clone Robotics	27	– fluidic muscle	<910	–	–	array	–	no	internal- only	0	2025-11-24	video
onex_neo_ hand_2026	1X Tech.	25	25 tendon	–	45	vendor, no protocol	unstated	–	no	an- nounced	0	2026-07-09	datasheet
leap_hand_ v2_adv_ 2025	CMU	21	17 tendon	–	–	–	none	3 000	yes	an- nounced	2	undated	datasheet
sanctuary_ phoenix_ hand_2024	Sanctuary AI (Sanctuary Cognitive...	21	– fluidic muscle	–	–	–	unstated	–	no	internal- only	0	2024-12-17	press
ilda_hand_ 2021	Korea Univ. Bicchi)	20	15 linkage	1 100	34	fingertip normal	array	–	no	prototype	0	2021-11-05	datasheet
pisa_iit_ softhand_ 2014		19	1 tendon	–	–	–	none	–	no	prototype	0	2014	datasheet
faive_ hand_2023	ETH Zurich	16	11 tendon	1 100	–	–	none	–	–	prototype	1	2023	datasheet
boston_ dynamics_ atlas_ hand_2026	Boston Dynamics	–	– geared	–	–	–	unstated	–	no	prototype	0	2026-01-05	press
daxo_ muscle_v0_ 2025	Daxo Robotics	–	– tendon	750	–	–	–	–	–	prototype	0	2026-09-16	press
figure_03_ hand_2025	Figure AI	–	–	–	–	–	unstated	–	no	prototype	0	undated	press
paxini_ dexh13_ 2024	PaXini Tech	–	–	–	–	–	array	–	–	an- nounced	0	2026-01-07	press
tesla_ optimus_ hand_2025	Tesla	–	– tendon	–	–	–	unstated	–	no	prototype	0	2023-12-13	video
xiaomi_ cyberone_ hand_2026	Xiaomi	–	– tendon	–	–	–	unstated	–	no	prototype	0	2026-03-30	press

TABLE III: 15 simulators and physics engines. 40 of the 105 cells in the 7 property columns (38%) are –: no source stated the value. *contact model* and *solver* are controlled vocabularies: *soft* for a compliant or regularised contact, *convex relaxation* for a convexified formulation, *complementarity* for an LCP, *hard NCP* for a nonlinear complementarity problem, *impulse* for a velocity-level update, and *rigid, model unstated* where the source describes collision geometry but names no contact law. The sentence each was read from runs to several hundred characters and is reproduced in full in Table XII, because Section IV’s argument turns on its wording. *penetration exposed* is whether the engine makes interpenetration depth readable by a user’s code without patching it. *hands shipped* counts first-party hand models in the parsed snapshot.

engine	contact model	solver	diff.	GPU	dt (s)	penetration exposed	hands shipped
brax_2021	convex relaxation	explicit/spring	yes	yes	–	no	1: synthetic 4-fingered claw hand
comfree_sim_2026	complementarity-free	closed form	–	yes	0.002	yes	–
dojo_2022	hard NCP	interior point	yes	no	–	yes	–
genesis_2024	soft	Newton	yes	yes	–	–	1: Shadow Hand
isaacgym_2021	convex relaxation	PGS	no	yes	0.008333	no	3: Shadow Dexterous Hand; Allegro Hand; TriFinger
isaacclab_2025	rigid, model unstated	Gauss-Seidel	no	yes	–	no	4: KUKA Allegro hand; ShadowHand; AbilityHand; Fourier...
maniskill13_2024	convex relaxation	PGS	–	yes	–	–	5: Allegro Hand; Ability Hand; Inspire Hand; Delto 3-finger...
mujoco_2012	complementarity	Newton	no	no	–	–	–
mujoco_convex_contact_2014	complementarity-free	closed form	no	no	0.01	–	–
mujoco_warp_2025	soft	PGS	no	yes	–	–	–
newton_2025	soft	primal-dual	–	yes	–	–	1: Allegro Hand
orbit_2023	convex relaxation	PGS	no	yes	–	no	1: Allegro hand
pybullet_2016	–	–	–	yes	–	–	–
raisim_2018	–	–	–	–	–	–	–
sapien_2020	convex relaxation	PGS	–	no	–	–	–

TABLE IV: Reward-term families across the 21 in-hand reorientation methods whose objective could be read term by term. A mark is **P** where the term is in the paper and either no code was released or it is absent from the released reward code, **C** where it is in the released code and not in the paper’s stated reward, **B** where it is in both, and **Z** where the released code carries the term with a weight of zero. 97 of the 189 term cells (51%) are blank, which means the method does not use that family; a blank here is not –, because the objective was read in full for every row in this table. The one –column is *mismatch*, which cannot be settled for a method that released no code. *terms* is the count of reward terms the paper states. Marks are read from the per-method source section recorded in `corpus/reward_matrix.json`.

method	yr	goal/rot	obj vel	fing-obj	hand pose	act rate	effort	drop	contact	bonus	terms	code	mismatch
openai_ dexterity_ 2018	2018	P						P		P	3	no	–
openai_ rubiks_cube_ 2019	2019	P						P		P	3	no	–
dextreme_ 2022	2022	B				B	B	Z		B	6	yes	yes
hora_2022	2022	B	B		B		B				5	yes	yes
visual_ dexterity_ 2022	2022	B		B	P		B	B	P	B	7	yes	yes
dexpbt_2023	2023	B		B		C	P	Z		B	4	yes	yes
eureka_2023	2023	P	P	P							–	yes	yes
robot_ synesthesia_ 2023	2023	P	P	P		P	P	P			6	no	–
rotateit_ 2023	2023	P	P		P		P				6	–	–
rotating_ without_ seeing_2023	2023	P	P	P			P	P			6	–	–
anyrotate_ 2024	2024	P	P		P		P	P	P	P	10	no	–
demostart_ 2024	2024									P	1	no	–
dreureka_ 2024	2024	P	P		P			P			4	yes	yes
penspin_2024	2024	B	B		B	Z	B				7	yes	yes
dexndm_2025	2025	P	P		P		P			P	7	no	–
dexremoe_ 2025	2025	P	P			P				P	5	no	–
dexteritygen_ 2025	2025										–	no	–
force_grasp_ sim2real_ 2026	2026	P			P	P	P	P	P	P	7	no	–
poise_2026	2026	P					P	P	P	P	4	no	–
teledexter_ 2026	2026	P								P	–	no	–
viserdex_ 2026	2026	P	P			P	P	P		P	10	no	–

TABLE V: 30 teleoperation and human-data systems, every corpus row that names an operator interface. Two columns rather than one: the interface and hand columns need about 140 pt between them before the hand names start breaking mid-word, and at that width six columns do not fit 252 pt. 60 of the 150 cells in the 5 columns other than the key (40%) are –: no source stated the value. *interface* is a controlled vocabulary. *latency* is whatever the source measured, reduced to its leading figure, and is – where a paper claims low latency without a number, which is the usual case. *rig USD* is the authors’ own bill of materials where they published one. *data collected* is the released trajectory count and hours, and a system built to be an interface rather than a dataset states neither.

system	interface	robot hand	latency	rig USD	data collected
dexpilot_2020	glove	Wonik Robotics Allegro hand	about one...	–	–
dexmv_2021	camera	Adroit Hand	–	–	700 traj, 7 h
dexvip_2022	motion capture	Adroit Hand	–	–	–
dime_2022	VR headset	Allegro Hand	30 Hz	–	30 traj
holo_dex_2022	VR headset	Allegro Hand	100 ms	399	–
videodex_2022	–	LEAP Hand	–	–	965 traj
anyteleop_2023	VR headset	Allegro Hand	35 ms	–	–
pgdm_2023	–	ShadowHand	–	–	40 traj
ace_teleop_2024	VR headset	Ability Hand	27 Hz	600	–
bidex_teleop_2024	VR headset	LEAP Hand	–	6 000	50 traj
bunny_visionpro_2024	VR headset	Ability Hand	3.43 ms	–	–
cyberdemo_2024	VR headset	Allegro Hand	–	–	–
dexcap_2024	glove	LEAP Hand	–	4 000	787 traj, 4.5 h
egomimic_2024	exoskeleton/leader	parallel-jaw gripper	–	1 000	2 150 traj, 4 h
hudor_2024	VR headset	Allegro Hand	–	–	4 traj
okami_2024	VR headset	Inspire Hand	–	–	–
open_television_2024	VR headset	Inspire Robots hand	60 Hz	–	–
dexteritygen_2025	glove	Allegro Hand	–	–	–
dexumi_2025	exoskeleton/leader	Inspire Hand	per-sensor...	–	1 355 traj
dexwild_2025	VR headset	LEAP Hand	–	–	9 290 traj, 46.2 h
doglove_2025	exoskeleton/leader	LEAP Hand	30 Hz	600	–
egozero_2025	glove	Franka Panda parallel-jaw gripper	–	–	700 traj
geometric_retargeting_2025	glove	Allegro Hand	1 kHz	–	–
h_rdt_2025	none	parallel-jaw	–	–	338k traj, 829 h
humanoid_policy_human_policy_2025	VR headset	Inspire Hand	–	700	–
wm_dex_human_videos_2025	–	Allegro Hand	–	–	829 h
dexteleop0_2026	VR headset	Sharpa Wave	30 Hz	–	–
egoscale_2026	glove	Sharpa Wave	–	–	20854 h
teledexter_2026	glove	SharpaWave	–	–	–
unidex_2026	VR headset	Inspire	–	–	52k traj

TABLE VI: The 20 most-mentioned method papers in the corpus, as a body version of the full method table. The full table is 112 rows and eleven columns and cannot be read in a body float at any type size, so it runs as Table X in the appendix with the algorithm, DoF, bimanual and unseen-object columns restored. Rows are ranked by whole-word mentions in the other parsed papers of papers/md, by the rule of Section VII: diffusion_policy_2023 70; openai_dexterity_2018 64; dexmv_2021 50; aloha_act_2023 49; dapg_2017 49; dextreme_2022 44; dexcap_2024 40; dexpilot_2020 39; umi_2024 36; anyteleop_2023 34; hora_2022 34; visual_dexterity_2022 32; openvla_2024 29; dime_2022 28; rotating_without_seeing_2023 27; unidexgrasp_2023 27; open_television_2024 26; pddm_2019 25; dp3_2024 24; hato_visuotactile_2024 23. A mention count is a count of mentions, not of use. 13 of the 140 cells in the 7 columns other than the key and the year (9%) are –: no source stated the value. *penetration* is *none* where the row’s note settles that the work does not address interpenetration, and – where no note settles it either way; no row in the corpus reports a penetration number for its own trained policy. *task* and *paradigm* print the first two tags of a row that carries more, marked with a trailing plus. *simulator* is normalised by the same rule tools/check_numbers.py uses, which folds Isaac Gym spellings together and keeps Isaac Lab apart from it.

method	yr	task	paradigm	hand	simulator	real trials	penetration	code
diffusion_policy_2023	2023	grasp, tool+	BC, diff	–	mujoco	20	none	yes
openai_dexterity_2018	2018	reorient	RL	ShadowRobot Dexterous Hand	mujoco	10	none	no
dexmv_2021	2021	grasp, tool	RL+demo, data	Adroit Hand	mujoco	–	none	yes
aloha_act_2023	2023	bi-coord	BC	parallel-jaw gripper	mujoco	25	none	yes
dapg_2017	2017	grasp, reorient+	RL+demo	ADROIT	mujoco	–	none	yes
dextreme_2022	2022	reorient	RL	Allegro Hand	isaacgym	10	none	yes
dexcap_2024	2024	grasp, bi-coord+	BC, diff+	LEAP Hand	none	60	–	yes
dexpilot_2020	2020	reorient, grasp+	teleop	Wonik Robotics Allegro hand	–	150	none	no
umi_2024	2024	tool, bi-coord+	BC, diff+	–	–	260	–	yes
anyteleop_2023	2023	grasp, tool+	teleop, RL+demo	Allegro Hand	sapien	100	none	yes
hora_2022	2022	reorient	RL, distil	Allegro Hand	isaacgym	240	none	yes
visual_dexterity_2022	2022	reorient	RL, distil	D’Claw	isaacgym	20	none	yes
openvla_2024	2024	other	VLA	–	libero...	230	none	yes
dime_2022	2022	reorient, grasp	teleop, BC+	Allegro Hand	mujoco	10	none	yes
rotating_without_seeing_2023	2023	reorient	RL	Allegro Hand	isaacgym	30	none	–
unidexgrasp_2023	2023	grasp	grasp-syn, RL+	ShadowHand	isaacgym	–	penalised	yes
open_television_2024	2024	bi-coord, tool	teleop, BC	Inspire Robots hand	none	20	none	yes
pddm_2019	2019	reorient, other	MPC	Shadow Hand	mujoco	–	none	yes
dp3_2024	2024	reorient, grasp+	BC, diff	Shadow Hand	isaacgym	40	none	yes
hato_visuotactile_2024	2024	handover, bi-coord	diff	Psyonic Ability Hand	–	10	none	yes

TABLE VII: The proposed evaluation protocol. Two columns rather than one: four columns of clauses do not fit 252pt without breaking words. No cell is –, because every cell here is a proposal rather than a reading of a source, and the table is the one in this paper that no corpus row can fill. The *how* and *why* columns of the long form, which carry the design of each measurement and the reason for each count, are in Table XIII; a paragraph of rationale does not belong in a body float. Every count is derived in Section VII, and the bill for two policies on three tasks is 342 matched plus 600 unseen real rollouts, against 1200 on the independent-arm count the paired design replaces.

axis	what is measured	minimum trials	reported with it
Task success	fraction of episodes meeting a criterion written before the run, and a graded rubric score in $[0, 1]$	100 real and 200 sim per policy, task and condition for an absolute rate at a 10-point interval; a comparison alone needs 57 matched pairs per arm	raw counts, the criterion verbatim, the rubric, the initial-condition protocol, and an interval whose kind is named
Robustness	success under each perturbation axis, and the ratio to the unperturbed anchor	40 per axis per policy to rank the axes; 101 per axis and 101 at the anchor before any claim that a named axis hurt	both absolute rates with their Wilson intervals, the axis ranges, and the ratio without an interval unless the certifying count was run
Unseen objects	mean over held-out objects of per-object success	20 objects at 5 trials to screen; 93 objects for a 10-point claim	the object list, the per-object rates, a bootstrap interval over objects, and the within-object trial count
Physical plausibility	maximum and mean hand-object penetration depth per rollout, and the fraction of frames above 2 mm	100 rollouts, simulation only	an interval over rollouts, the sample density, the threshold, the solver’s depenetration settings, and at least one rendered rollout
Sample and wall-clock cost	environment steps and wall-clock to the checkpoint that produced the headline number	3 training seeds	the range over seeds, and the statement that 3 seeds is a range and not an interval
Real-robot transfer	paired real and simulated outcomes on matched initial conditions, their correlation, and the rectifier variance against the real variance	the 100 real trials paired to 100 sim; about 457 pairs to separate two correlations 0.1 apart	the correlation with its Fisher-z interval, the variance ratio with its bootstrap interval, and which decision the count supports
Reproducibility	the released config and checkpoints against the paper’s stated objective	not a trial count	the file and line of every disagreement, or an explicit statement that none was found

TABLE VIII: The matrix, for someone else to fill. 12 rows and 84 cells, all 84 of them empty. An empty cell here is not –: it is a measurement nobody has made, which is the point of the table, and no cell in it has a source to be missing. The first row is a worked example and every number in it is fabricated. Rows are the 12 most-mentioned dexterous-hand policy methods in the corpus, by the rule of Section VII, scored on whole-word matches over papers/md: openai_dexterity_2018 64; dexmv_2021 50; dapg_2017 49; dextreme_2022 44; dexcap_2024 40; hora_2022 34; visual_dexterity_2022 32; rotating_without_seeing_2023 27; unidexgrasp_2023 27; pddm_2019 25; hato_visuotactile_2024 23; dexpoint_2022 22. A mention count is a count of mentions, not of use. * marks a work matched on its title rather than on a short name; a title is matched mostly inside reference lists and a short name in running text, so the two kinds of count are not comparable. Each column is an axis of Table VII and is filled in the units that table asks for.

method	task success	robustness	unseen objects	plausibility	cost	transfer	reproducibility
<i>worked example</i>	0.72 ± 0.09 (100)	lighting 0.41 (40), ratio 0.57	0.55 [0.41, 0.68], 20 obj	3.1 / 0.8 mm, 0.12	1.2e9 steps, 46 GPU-h	0.61 [0.47, 0.72]	train. yaml:88
openai_ dexterity_ 2018*							
dexmv_2021							
dapg_2017*							
dextreme_2022							
dexcap_2024							
hora_2022*							
visual_ dexterity_ 2022*							
rotating_ without_ seeing_2023*							
unidexgrasp_ 2023							
pddm_2019*							
hato_ visuotactile_ 2024*							
dexpoint_2022							

TABLE IX: 14 existing surveys, one per corpus row of class `survey`, newest first. 3 could not be obtained and 1 was fetched too late to be read into a note; those rows say so in every column rather than leaving blanks, because a blank row would read as a survey that covers nothing. Of the remaining cells, 0 print `–`, which here means the quotation that supported the cell has gone from `papers/notes/` and the claim is no longer checkable rather than that no source stated it. Every filled cell is read from `papers/notes/<key>.md` and is checked against a quotation from that note by `tools/make_survey_table.py`. Cells are cut at a word boundary; the *taxonomy* and *gaps* columns, and every cell at full length, are in Table XIV. The columns record what each work covers, not how well, and a blank in *bimanual* or *hardware* is a scope decision by its authors rather than a failure.

survey	yr	scope	bimanual covered	hardware covered	evaluation covered
isaac_sim_2026	2026	one simulator’s ecosystem and application domains, reviewed rather...	one cited GR00T task called bimanual, no hand, DoF or number	no hand named anywhere in paper or code parse	none. The paper runs no experiment of its own
zhao_dexhand_survey_2026	2026	dexterous hands end to end: hardware anatomy, methods, datasets,...	one subsection, III-F, 16 cited works, no coordination analysis	Table I, 29 hands, 12 columns, secondary values	names two layers: physical plausibility including penetration before...
an_dexil_survey_2025	2025	imitation learning for multi-fingered end-effectors	one subsection, II.E, framed as multi-agent	hands named in prose, no hand table	no metric defined and no trial count. Calls for protocols, proposes...
bai_unified_manip_survey_2025	2025	all of robot manipulation. Dexterous manipulation is one of ten task...	bimanual means two arms. One dual-hand mention in the paper	hands named, no DoF or actuation table	success rate and checkpoint selection, six lines. Never says how...
welte_iil_survey_2025	2025	interactive imitation learning, seven dexterous works found	three mentions in 687 lines, no section	Table 1, 15 commercial hands, manufacturer figures	no metric defined, no benchmark table
nine_physics_engines_review_2024	2024	nine physics engines for RL research, scored on documentation and...	none. MARL readiness is the multi-agent axis	none. Ant and humanoid RL bodies are the running examples	no benchmark of its own. Throughput claims are second-hand
contact_models_comparison_2023	2023	LCP, CCP, RaiSim and NCP contact models re-implemented in one...	none	one Allegro hand as a benchmark system, a ball dropped into it	NCP criterion, self-consistency against a 1e-5 s reference, iteration...
firoozi_foundation_models_2023	2023	foundation models in robot decision-making, perception and embodied AI	none, zero occurrences	none. Parallel-jaw end effectors throughout	none defined. Benchmarking appears as a reproducibility problem
zhao_sim2real_survey_2020	2020	sim-to-real transfer in deep RL, eight pages, 21 works tabulated	none, zero occurrences	two cited hand works, no hand table	no metric defined, no trial count, no success rate
piazza_century_2019	2019	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>
physics_engine_comparison_2015	2015	five engines on one shared model: speed, self-consistency,...	none	one 35-DoF rig modelled on the Shadow Hand	largest timestep that holds a grasp, and a speed-accuracy Pareto curve
roa_suarez_grasp_quality_2015	2015	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>
ma_dollar_dexterity_2011	2011	<i>PDF fetched 2026-09-18 after the note was written, no note read from...</i>	<i>PDF fetched 2026-09-18 after the note was written, no note read from...</i>	<i>PDF fetched 2026-09-18 after the note was written, no note read from...</i>	<i>PDF fetched 2026-09-18 after the note was written, no note read from...</i>
okamura_overview_2000	2000	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>	<i>source not obtained</i>

TABLE X: Every method row in the corpus, 112 of them, in the long form the body version of Table VI cuts down. Run across both columns in one-column mode because fourteen columns do not fit a two-column float. 224 of the 1344 cells in the 12 columns other than the key and the year (17%) are –: no source stated the value. The algorithm, hand and simulator columns are the corpus strings, cut at a word boundary where a cell would otherwise run past its column; `corpus/rows/` holds every field at full length.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetration	code
ferrari_canny_1992	1992	–	–	–	–	–	–	–	–	–	–	–	no
dapg_2017	2017	grasp, reorient, tool+	RL+demo	NPG (Natural Policy Gradient) + BC pretraining (DAPG)	ADROIT	24	no	MuJoCo	no	–	–	none	yes
openai_dexterity_2018	2018	reorient	RL	PPO	ShadowRobot Dexterous Hand (EDC electric-motor version)	24	no	MuJoCo	yes	10	–	none	no
openai_rubiks_cube_2019	2019	reorient	RL, distil	PPO with Automatic Domain Randomization (ADR); Policy Cloning (DAGGER/distillation-style) for architecture...	Shadow Dexterous E Series Hand (E3M5R)	20	no	MuJoCo (+...)	yes	10	–	none	no
pddm_2019	2019	reorient, other	MPC	PDDM: ensemble of feed-forward dynamics models + MPPI-style filtering/reward-weighted-refinement trajectory...	Shadow Hand (24-DoF, real and sim); D'Claw (9-DoF, sim...)	–	no	MuJoCo	yes	–	–	none	yes
dexpilot_2020	2020	reorient, grasp, tool+	teleop	DART model-based hand tracker bootstrapped by two learned neural priors (GloveNet ResNet-50 keypoint...)	Wonik Robotics Allegro hand (16-DoF, 4x4-joint fingers),...	16	no	–	yes	150	–	none	no
castro_sap_contact_2021	2021	bi-coord	–	SAP (Semi-Analytic Primal) solver: a custom Newton-type solver on an unconstrained convex primal...	Allegro hand (16 DoF each), two	16	yes	Drake	no	–	–	measured	yes
dexmv_2021	2021	grasp, tool	RL+demo, data	DAPG / SOIL / GAIL+ on top of TRPO, using demonstrations translated from human video via TSV retargeting +...	Adroit Hand	30	no	MuJoCo	no	–	500	none	yes
dexpoint_2022	2022	grasp, other	RL	PPO	Allegro Hand	16	no	SAPIEN	yes	260	40	none	yes
dextreme_2022	2022	reorient	RL	PPO (rl-games implementation, LSTM actor/critic, asymmetric/state-privileged critic)	Allegro Hand	16	no	Isaac Gym (PhysX)	yes	10	–	none	yes
dexvip_2022	2022	grasp	RL	PPO with a video-mined consensus grasp-pose auxiliary reward (R_pose) combined with an affordance reward...	Adroit Hand	30	no	MuJoCo	no	–	–	none	no
dime_2022	2022	reorient, grasp	teleop, BC, RL+demo	INN/VINN (nearest-neighbor imitation) vs. BC; DAPG/PPO/BCRL (simulation RL-finetuning)	Allegro Hand	–	no	MuJoCo	yes	10	–	none	yes
holo_dex_2022	2022	reorient, grasp, tool	teleop, BC	VINN (BYOL nearest-neighbor) vs. Behavior Cloning / BC-Rep	Allegro Hand	16	no	–	yes	10	10	none	yes
hora_2022	2022	reorient	RL, distil	PPO	Allegro Hand (Wonik Robotics)	16	no	IsaacGym	yes	240	30	none	yes
mjpc_2022	2022	reorient	MPC	Predictive Sampling: zero-order, derivative-free sampling-based shooting MPC over a spline-parameterised...	Shadow Hand (MuJoCo Menagerie model)	–	no	MuJoCo	no	–	–	none	yes
pang_global_planning_2022	2022	reorient, grasp, bi-coord+	trajopt	iMPC (iLQR-style trajectory optimizer over a locally-smoothed convex quasi-dynamic contact model, CQDC) for...	Allegro Hand (full 3D model, in-hand rotation / pen...)	–	yes	Custom Convex...	yes	–	–	constrained	yes
videodex_2022	2022	grasp, reorient, tool	BC, data	two-stream Neural Dynamic Policy (NDP) trajectory regression (L1 loss): pretrained on human-video-retargeted...	LEAP Hand	16	no	–	yes	–	–	none	no
visual_dexterity_2022	2022	reorient	RL, distil	PPO (teacher, privileged state) + DAGger imitation distillation to a depth-point-cloud student (two-stage:...	D'Claw (open-source, low-cost; 3-finger and modified...	–	no	Isaac Gym (PhysX)	yes	20	12	none	yes

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Table X, continued.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetration	code
aloha_act_2023	2023	bi-coord	BC	ACT (Action Chunking with Transformers, CVAE)	parallel-jaw gripper (not dexterous), two	–	yes	MuJoCo	yes	25	–	none	yes
anyteleop_2023	2023	grasp, tool, handover	teleop, RL+demo	DAPG (downstream IL); keypoint-vector retargeting optimization (dex_retargeting)	Allegro Hand	–	no	SAPIEN	yes	100	–	none	yes
artigrasp_2023	2023	grasp, tool, bi-coord	RL	PPO (D-Grasp-based implementation)	MANO hand model, two	51	yes	RaiSim	no	–	1	none	yes
dexdeform_2023	2023	bi-coord, other	BC, trajopt	latent skill model (VAE over demonstrations) for skill-planned trajectory initialization, refined by...	Shadow Dexterous Hand (simulated)	28	yes	Plas-ticineLab...	no	–	–	none	yes
dexpbt_2023	2023	grasp, reorient, bi-coord	RL	PPO (rl_games) with decentralized Population-Based Training (PBT) over RL hyperparameters and reward-shaping...	Allegro Hand	16	yes	Isaac Gym (PhysX)	no	–	–	none	yes
dexterous_functional_grasping_2023	2023	tool	RL	PPO	LEAP hand	16	no	IsaacGym	yes	70	5	–	no
diffusion_policy_2023	2023	grasp, tool, bi-coord+	BC, diff	conditional DDPM over action sequences (CNN or transformer noise-prediction network), DDIM at inference	–	–	yes	task-specific:...	yes	20	–	none	yes
dynamic_handover_2023	2023	handover, bi-coord	RL	MAPPO	Allegro Hand	16	yes	IsaacGym	yes	15	14	none	no
eureka_2023	2023	reorient, bi-coord, other	RL	PPO with LLM-authored (GPT-4) evolutionary reward search	Shadow Hand (also Allegro Hand; bimanual pairs of Shadow...	20	yes	IsaacGym	yes	–	–	none	yes
pgdm_2023	2023	grasp, tool	RL	PPO (Stable-Baselines3) with an object-trajectory-tracking reward plus pre-grasp-based exploration...	ShadowHand	24	no	MuJoCo	yes	–	–	none	no
physhoi_2023	2023	track-ref	RL	PPO	SMPL-X humanoid (simulated, not a robot hand)	90	no	Isaac Gym	no	–	–	none	yes
robot_synesthesia_2023	2023	reorient	RL, distil	PPO	Allegro Hand	16	no	Isaac Gym	yes	5	–	none	no
rotateit_2023	2023	reorient	RL, distil	PPO	AllegroHand (Wonik Robotics)	16	no	IsaacGym	yes	–	15	none	–
rotating_without_seeing_2023	2023	reorient	RL	PPO	Allegro Hand	16	no	IsaacGym	yes	30	5	none	–
unidexgrasp_2023	2023	grasp	grasp-syn, RL, distil	PPO (teacher) + DAgger (student distillation)	ShadowHand	26	no	Isaac Gym	no	–	241	penalised	yes
unidexgrasp_pp_2023	2023	grasp	RL, distil	PPO + DAgger (with critic distillation)	Shadow Hand	24	no	Isaac Gym	no	–	–	none	yes
ace_teleop_2024	2024	grasp, tool	teleop, BC	3D Diffusion Policy (DP3) for xArm platforms; ACT for H1 humanoid (downstream IL); IK-based scale/recenter...	Ability Hand / Inspire Hand / parallel-jaw gripper...	–	yes	–	yes	10	18	none	no
anyrotate_2024	2024	reorient	RL, distil	PPO	Allegro Hand	16	no	IsaacGym	yes	–	10	none	no
asymdex_2024	2024	bi-coord, grasp	RL	PPO	Shadow Hand (30-DoF: 24-DoF hand + 6-DoF floating wrist) in...	30	yes	NVIDIA Isaac Gym	yes	20	–	none	yes
bidex_teleop_2024	2024	bi-coord, handover, tool	teleop, BC	ACT (Action Chunking Transformer)	LEAP Hand (16 DoF); LEAP Hand V2 (21 DoF) used for 'extreme...	16	yes	none	yes	20	–	none	no
bidexhd_2024	2024	tool, bi-coord	RL, distil	IPPO (teacher) + DAgger (student distillation)	LEAP Hand, left and right	16	yes	Isaac Gym (PhysX)	no	–	–	none	yes
bimangrasp_2024	2024	grasp, bi-coord	grasp-syn, diff	MALA (Metropolis-adjusted Langevin) energy optimisation + DDPM (BimanGrasp-DDPM)	Shadow Hand, pair	22	yes	Isaac Gym	no	–	225	penalised	no
bunny_visionpro_2024	2024	grasp, tool, bi-coord	teleop, BC	ACT, Diffusion Policy, DP3 (downstream IL); SQP fingertip-vector hand retargeting + unified...	Ability Hand	24	yes	–	yes	10	–	none	yes

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Table X, continued.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetration	code
cyberdemo_2024	2024	grasp, reorient, tool	BC, teleop, data	ACT (Action Chunking with Transformers) BC policy, trained on sim teleoperation demonstrations augmented via...	Allegro Hand	16	no	SAPIEN	yes	20	2	none	no
demostart_2024	2024	grasp, tool, reorient+	RL, distil	MPO (teacher, auto-curriculum) + BC distillation (student, Perceiver-Actor-Critic)	DEX-EE Hand	12	no	MuJoCo	yes	100	–	none	no
dexcap_2024	2024	grasp, bi-coord, tool	BC, diff, data	Diffusion Policy with Perceiver point-cloud encoder (DP-perc)	LEAP Hand	16	yes	none	yes	60	9	–	yes
dexmimicgen_2024	2024	bi-coord, grasp, tool	data, BC	MimicGen-style per-arm subtask transform/replay generation; downstream BC-RNN / BC-RNN-GMM / Diffusion Policy	Inspire dexterous hand (6-DoF, real-world GR1); unnamed...	6	yes	RoboSuite (MuJoCo)	yes	20	–	–	yes
dp3_2024	2024	reorient, grasp, tool+	BC, diff	Diffusion Policy backbone (DDPM training / DDIM inference) conditioned on a compact 3D point-cloud feature...	Shadow Hand (Adroit, Bi-DexHands, DexDeform, DexMV...)	–	yes	MuJoCo (Adroit,...)	yes	40	5	none	yes
dreureka_2024	2024	reorient	RL	PPO with LLM-authored (GPT-4) reward search + reward-aware physics prior domain randomization	LEAP hand	16	no	Isaac Gym	yes	–	–	none	yes
egomimic_2024	2024	grasp, tool, bi-coord	BC	shared-backbone ACT variant co-trained on paired egocentric human video and teleoperated bimanual robot...	parallel-jaw gripper (not a dexterous hand; see note scope...)	1	yes	–	yes	135	–	none	yes
graspxl_2024	2024	grasp	RL	PPO	MANO (also Shadow Hand, Allegro Hand, Faive Hand)	45	no	RaiSim	no	–	503409	none	yes
hato_visuotactile_2024	2024	handover, bi-coord	diff	Diffusion Policy (DDPM, CNN-based)	Psyonic Ability Hand, two	6	yes	–	yes	10	–	none	yes
hudor_2024	2024	grasp, tool	RL	open-loop IK replay of a single retargeted human fingertip trajectory plus online residual RL (DrQv2) trained...	Allegro Hand	16	no	–	yes	10	–	none	no
humanplus_2024	2024	track-ref, loco	RL, BC	PPO (HST shadowing policy) + decoder-only Transformer BC (HIT)	Inspire-Robots RH56DFX, two	6	yes	Isaac Gym-based...	yes	10	–	none	yes
objdex_2024	2024	track-ref, bi-coord	BC, RL, distil	BC (Transformer high-level planner) + PPO (low-level controller)	Shadow Hand	–	yes	–	yes	80	1	none	no
okami_2024	2024	grasp, tool, bi-coord	trajopt, BC	single-video imitation: SLAHMR-extended SMPL-H body+hand reconstruction, GPT-4V/Grounded-SAM/Cutie...	Inspire Hand (x2)	6	yes	RoboSuite (2 of 6...)	yes	12	–	none	yes
omnigrasp_2024	2024	grasp, track-ref	RL, distil	PPO (hierarchical: PHC-X finger imitator → PULSE-X distilled latent prior → PPO over latent action space)	SMPL-X whole-body humanoid (simulated, not a robot hand)	90	yes	Isaac Gym	no	–	5	none	yes
omnih2o_2024	2024	track-ref, loco	RL, distil	PPO (teacher) + DAgger (student) + diffusion policy (autonomy layer)	Inspire hands (open-loop, DoF not stated)	–	yes	–	yes	20	–	–	yes
open_television_2024	2024	bi-coord, tool	teleop, BC	ACT (Action Chunking Transformer) with DinoV2 ViT backbone	Inspire Robots hand (6 actuated DoF, 12 total); Fourier...	6	yes	none	yes	20	–	none	yes
openvla_2024	2024	other	VLA	autoregressive next-token prediction over discretized (256-bin per-dimension, quantile-based) action tokens...	–	–	no	LIBERO benchmark...	yes	230	–	none	yes
penspin_2024	2024	reorient	RL, distil	PPO + BC	Allegro Hand	16	no	Isaac Gym	yes	50	7	none	yes
pi0_2024	2024	other	VLA, flow	conditional flow matching (linear-Gaussian/optimal-transport probability path), forward-Euler integration at...	–	–	yes	–	yes	10	–	none	yes

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Table X, continued.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetra- tion	code
pianomime_ 2024	2024	music	RL+demo, distil	PPO (per-song specialists); DDPM diffusion (generalist)	Shadow Hand E3M5	23	yes	ROBOPI- ANIST (MuJoCo)	no	–	–	none	yes
rdt1b_2024	2024	bi-coord	diff	RDT (DiT + cross-attention, QKNorm/RMSNorm, MLP decoder, Alternating Condition Injection)	parallel-jaw gripper (ALOHA-style, Cobot Mobile ALOHA), two	–	yes	–	yes	139	2	none	yes
resdex_2024	2024	grasp	RL, distil	PPO + DAgger	ShadowHand	18	no	IsaacGym	no	–	241	none	yes
twisting_ lids_2024	2024	tool, bi-coord	RL	PPO (asymmetric actor-critic)	Allegro Hand, two	16	yes	Isaac Gym	yes	20	15	none	no
umi_2024	2024	tool, bi-coord, other	BC, diff, data+	Diffusion Policy (unmodified loss/decoder); contribution is the hand-held data-collection hardware and the...	–	–	yes	–	yes	260	–	–	yes
articulated_ tools_ inhand_2025	2025	tool	RL, BC, distil	PPO (oracle) + BC (student + CATFA online adaptation)	Inspire Hand	6	no	IsaacLab	yes	50	–	none	no
being_h0_ 2025	2025	grasp, tool	VLA, BC	autoregressive next-token prediction over discretized MANO motion tokens (Grouped Residual Quantization) for...	6-DoF Inspire hand mounted on a 7-DoF Franka Research 3 arm...	6	no	–	yes	20	–	none	yes
clutterdexgrasp_ 2025	2025	grasp	RL, diff, distil	PPO (teacher) + DP3 diffusion policy (student)	AgiBot dexterous hand	6	no	Isaac Gym	yes	167	2029	con- strained	no
cross_ embodiment_ world_models_ 2025	2025	other	world, MPC	DPI-Net (graph neural network particle-dynamics world model) + sampling-based receding-horizon MPC	PSYONIC Ability Hand (6-DoF) and Robot Era XHand (12-DoF)...	–	no	SAPIEN (rigid- body...	yes	20	–	none	no
dexgraspvla_ 2025	2025	grasp, other	VLA, diff	hierarchical VLA: frozen VLM high-level planner (bounding-box affordance) + DiT diffusion-policy controller...	PsiBot G0-R, 6-DoF, single (right) hand	6	no	–	yes	1287	360	none	yes
dexmachina_ 2025	2025	track-ref, bi-coord	RL	PPO (rl-games)	six URDFs: Inspire, Allegro, XHand, Schunk (main), Ability...	–	yes	Genesis	no	–	–	con- strained	yes
dexman_2025	2025	track-ref, bi-coord	RL, trajopt	PPO (RL_GAMES) residual policy on top of IK-retargeted human motion	Shadow Dexterous Hand	–	yes	NVIDIA Isaac Gym	no	–	–	–	no
dexndm_2025	2025	reorient	RL, distil	PPO (oracle) + BC (generalist) + supervised residual policy	LEAP hand	–	no	Isaac Gym	yes	–	–	none	no
dexplore_ 2025	2025	track-ref	RL, distil	PPO (teacher) + DAgger-style VAE distillation (student)	Inspire hand (also Allegro hand)	6	no	Isaac Gym	yes	–	2	none	yes
dexremoe_ 2025	2025	reorient	RL	PPO	GX11 three-fingered dexterous hand (custom)	11	no	IsaacGym	no	–	50	none	no
dexteritygen_ 2025	2025	reorient, tool, grasp	RL, diff	RL-generated Anygrasp-to-Anygrasp transitions used to train a UNet DDPM diffusion foundation controller...	Allegro Hand	16	no	–	yes	240	–	–	no
dexterous_ handover_ 2025	2025	handover, grasp	RL	PPO	Allegro Hand	–	no	IsaacLab	no	–	3	none	–
dextrack_ 2025	2025	track-ref	RL+demo	PPO + IL action-supervision loss	Allegro hand (sim); LEAP hand (real)	16	no	Isaac Gym	yes	–	–	measured	yes
dexumi_2025	2025	grasp, tool	BC, data	diffusion policy (DDPM-style action-chunking) trained on exoskeleton-collected demonstrations converted to...	Inspire Hand (12 DoF, 6 active) and XHand (12 active DoF)	–	no	–	yes	20	–	–	yes
dexvla_2025	2025	grasp, tool, bi-coord	VLA, diff	ScaleDP (Scale Diffusion Policy, transformer-based diffusion action head, up to 1B parameters, multi-head per...	Robotiq parallel-jaw gripper (Franka rig a, Bimanual UR5e...	–	–	LIBERO (sec- ondary...	yes	10	30	none	yes
dexwild_2025	2025	grasp, tool, bi-coord	BC, data	diffusion U-Net policy (also compared against ACT) co-trained on human wearable-rig demonstrations and...	LEAP Hand / LEAP Hand V2 Advanced	17	yes	–	yes	–	11	–	yes

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Table X, continued.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetration	code
dydexhandover_2025	2025	handover, bi-coord	RL	MAPPO (human-regularized, CTDE, with hybrid advantage estimation)	11-DoF (6 actuated) dexterous hand, per side (vendor not...)	11	yes	NVIDIA Isaac Sim /...	no	–	9	none	no
egozero_2025	2025	grasp, tool	BC	closed-loop Transformer policy (BC, Gaussian NLL loss) over a morphology-agnostic 3D-point state-action space...	Franka Panda parallel-jaw gripper (not a dexterous hand);...	1	no	–	yes	15	–	none	yes
gemini_robotics_15_2025	2025	other	VLA	–	ALOHA parallel gripper (by platform convention, not...)	–	yes	MuJoCo (used to...)	yes	–	–	none	no
gemini_robotics_2025	2025	other, handover	VLA, distil	–	ALOHA 2 parallel gripper, two fingers (primary embodiment);...	–	yes	–	yes	20	–	none	no
geometric_retartargeting_2025	2025	grasp	teleop	per-finger MLP retargeting network (GeoRT) trained offline against five geometric losses: motion-direction...	Allegro Hand	–	no	–	yes	–	–	none	yes
gr_dexter_2025	2025	bi-coord, grasp	VLA	GR-Dexter (Mixture-of-Transformer VLA, flow matching + next-token prediction, following GR-3)	ByteDexter V2, two	21	yes	–	yes	125	23	none	no
groot_n16_2025	2025	grasp, tool, bi-coord+	VLA	flow-matching DiT action head (32 layers, per page and cross-confirmed by code changelog) fed by a Cosmos-2B...	–	–	yes	Galaxea R1 Pro...	yes	–	–	none	yes
groot_n1_2025	2025	grasp, tool, bi-coord+	VLA, flow	dual-system architecture: Eagle-2 VLM (System 2, reasoning) feeding a DiT flow-matching action head (System...	Fourier dexterous hands, on the Fourier GR-1 humanoid;...	–	yes	RoboCasa (Kitchen and...	yes	10	5	none	yes
h_rdt_2025	2025	grasp, tool, bi-coord	flow	2B-parameter flow-matching diffusion transformer (H-RDT), pretrained on EgoDex human hand-pose trajectories...	parallel-jaw / 2-jaw grippers at deployment (Aloha-Agilex,...	–	yes	RoboTwin 2.0	yes	25	–	none	no
helix_2025	2025	grasp, tool, handover	VLA	dual-network "System 1, System 2" VLA: S2 (7B VLM, 7-9Hz) produces a continuous latent conditioning vector...	–	–	yes	–	yes	–	–	none	no
human2sim2robot_2025	2025	track-ref	RL	PPO	Allegro Hand	16	no	IsaacGym	yes	70	–	none	yes
humanoid_policy_human_policy_2025	2025	grasp, tool	BC	Human Action Transformer (HAT), an ACT-style action-chunking transformer co-trained jointly on unified human...	Inspire Hand (x2, 5-fingered)	6	yes	–	yes	230	–	none	yes
humanoid_sim2real_recipe_2025	2025	grasp, bi-coord, handover	RL, distil	PPO (asymmetric actor-critic) for per-task specialist policies, distilled into a generalist via Diffusion...	Fourier GR1 hand (6 actuated + 5 underactuated DoF);...	6	yes	NVIDIA Isaac Gym	yes	10	–	–	no
maniptrans_2025	2025	track-ref, bi-coord	RL, BC, diff	PPO (two-stage: frozen generalist imitator + per-task residual policy)	Inspire Hand (12-DoF sim); also Shadow(22), MANO(22),...	12	yes	Isaac Gym	yes	–	–	none	yes
metis_2025	2025	grasp, tool, bi-coord	VLA	Autoregressive next-token cross-entropy over discretized 'motion-aware dynamics' tokens (VQ-VAE visual...	Inspire dexterous hand (paired, main real-robot eval);...	6	yes	–	yes	20	–	none	no
pi05_2025	2025	other	VLA, flow	hierarchical high-level discrete-subtask prediction (FAST tokenizer, autoregressive) + low-level...	–	–	yes	–	yes	40	–	none	yes

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Table X, continued.

method	yr	task	paradigm	algorithm	hand	DoF	bi	simulator	real	trials	unseen	penetration	code
pistar06_2025	2025	other	VLA, RL, flow	RECAP: advantage-conditioned policy extraction on a flow-matching VLA, using a distributional value function...	–	–	yes	–	yes	750	–	none	yes
wm_dex_human_videos_2025	2025	grasp	world, MPC	DexWM: a deterministic latent-space (DINOv2) world model with a CDiT-based predictor conditioned on MANO...	Allegro Hand	–	no	RoboCasa	yes	12	–	–	no
being_h05_2026	2026	bi-coord, other	VLA, flow	Rectified Flow (continuous action velocity-field prediction) + Masked Motion Token Prediction (discrete)...	Inspire Hand (Franka FR3); LinkerBot O6 (Unitree G1);...	6	–	LIBERO, RoboCasa...	yes	20	–	none	yes
bidexgrasp_2026	2026	grasp, bi-coord	grasp-syn, diff	bi-level QP-ADMM optimisation (region-pair init + decoupled per-hand force closure) + DDPM generator	Shadow Hand pair (sim); Inspire and BrainCo hands (real)	–	yes	MuJoCo	yes	260	30	measured	no
deximit_2026	2026	bi-coord, grasp, tool	data, BC, grasp-syn+	4D hand-object reconstruction (ST2+FPose) + LLM-based (Qwen3-VL) subtask decomposition with Action-Centric...	XHands (real-world deployment); exact simulation hand not...	–	yes	–	yes	–	–	constrained	no
dexora_2026	2026	grasp, tool, bi-coord	VLA, diff, teleop	Decoder-only diffusion transformer (28 layers, hidden size 1024, 16 attention heads) with T5 (language) and...	XHAND dexterous hand (paired), 12 fully actuated joints per...	12	yes	MuJoCo (digital twin)...	yes	20	–	none	–
dexteleop0_2026	2026	grasp, tool, bi-coord	teleop, MPC	box-constrained QP shared-autonomy residual controller (force tracking + multi-contact force-torque balance +...	Sharpa Wave (x2)	22	yes	NVIDIA IsaacSim 4.5	yes	35	–	–	no
egoscale_2026	2026	grasp, tool, bi-coord	flow, VLA	flow-based VLA (GR00T N1-style): VLM backbone + DiT action expert trained with a flow-matching objective, in...	Sharpa Wave (22-DoF); cross-embodiment target Unitree G1...	22	yes	–	yes	10	–	–	no
force_grasp_sim2real_2026	2026	grasp, reorient	RL	PPO (asymmetric actor-critic)	xHand	12	no	IsaacLab	yes	70	2	–	no
poise_2026	2026	reorient	RL	asymmetric PPO	Sharpa Wave hand	22	no	Isaac Lab	yes	10	1	none	no
simtoolreal_2026	2026	tool	RL	SAPG (PPO variant)	Sharpa five-fingered hand	22	no	IsaacGym	yes	120	12	none	no
teledexter_2026	2026	reorient, tool	RL, teleop	single-stage RL (SAPG) with consecutive subgoal co-tracking reward	SharpaWave (22-DoF, headline); also LeapHand (16-DoF)	22	no	Isaac Gym	yes	15	–	penalised	no
toporetaret_2026	2026	track-ref	trajopt, RL	constrained Laplacian-optimization retargeting + PPO tracking	Wuji Hand	–	no	–	yes	–	–	constrained	no
unidex_2026	2026	grasp, tool	flow, VLA, data	UniDex-VLA: a pi0-style flow-matching VLA with a Uni3D pointcloud encoder and Gemma-based backbone,...	Inspire, Leap, Shadow, Allegro, Ability, Oymotion, XHand,...	–	no	–	yes	20	1	–	no
viserdex_2026	2026	reorient	RL, distil	PPO (RSL-RL)	Allegro Hand	16	no	Isaac Lab	yes	50	–	none	no

TABLE XI: The sentences behind the controlled vocabularies of Tables I and II, for all 33 hands. 51 of the 198 cells in the 6 description columns (26%) are –: no source stated the value. *caveat on the row* is the `spec_caveat` field, which is where a figure with no reachable source is named as such; it is empty for a row that carries no caveat rather than for a row nobody checked. Cells are cut at a word boundary where they would otherwise run past the page; `corpus/rows/` holds every field at full length.

hand key	maker, in full	actuation, as the source describes it	what the force figure is	tactile, as the source describes it	date of the claim	caveat on the row
clone_robotics_hand_2024	Clone Robotics	hydraulic artificial muscle ('Myofiber' — small water-filled tubes that contract when pressurized), carbon-fiber bones and ligament-style tethers,...	–	pressure pads in the palm detect grip firmness (count not given); 70 inertial sensors track angle/speed (proprioception, not...	press, 2025-11-24, on a video of 2025-11-15	'under 2 pounds' is an upper bound, and the source does not say whether the 500 W pump and 36 valves are...
onex_neo_hand_2026	IX Technologies	quasi-direct-drive tendons via the 'IX Tendon Drive' at low gear ratios (~5:1-15:1); motors in the forearm, tendons pulled through the wrist	distal flexion force, peak vendor claim	high-resolution tactile sensing across fingertips and finger surfaces (normal force, contact location, shear); sensor...	vendor page, 2026-07-09	–
shadow_dexterous_hand_2005	Shadow Robot Company	tendon-driven, motors in the forearm (20 Smart Motor nodes, Maxon motors, PWM)	–	Shadow Tactile Fingertips (STF): 17x3 DoF Hall-effect taxels, 1000 Hz, uncalibrated; up to 5 fingertips; software also supports...	specification dated 2024-12-04	the four distal joints are coupled and not independently controllable, so the '24 DoF' quoted in learning...
proception_prohand_2026	Proception Inc (YC W25)	tendon-driven, motors pull cables to move the fingers	–	integrated skin-like sensors detecting contact, supporting grip control; same 'sensor skin' used on the companion ProGlove;...	launch post and press, 2026-06-29	the evidence is a shipping announcement, not a datasheet: no weight, force, actuated-DoF count or control...
sharpa_wave_2026	Sharpa Robotics (Sharpa – Pte Ltd)	–	vendor spec table with two unlabelled columns, probably two configurations; grip force is listed separately...	'Dynamic Tactile Array' (DTA), camera-type array at the fingertip: resolution 240x240 60x60, force resolution 0.02 N 0.05 N,...	undated page, copyright 2026, fetched 2026-09-17	the page footnotes 'Specifications may vary between products', and no column of the spec table is labelled
bidexhand_2025	Zhengyang Kris Weng, Center for Robotics and Biosystems, Northwestern University	cable-driven, N-configuration tendon routing (FeeTech servos, endless-loop antagonistic pull-pull); thumb CMC flexion driven via a 4-bar linkage...	fingertip normal force, calibrated sensor, five trials per finger, average peak	none mentioned	paper, 2025 (ICRA Dexterity workshop abstract)	the paper states 16 independently actuated DoF and 21 joints; the README states 15 servos driving 15 joints
leap_hand_v2_adv_2025	Kenneth Shaw, Deepak Pathak, Carnegie Mellon University	tendon (PIP/DIP coupled by a single tendon per finger); motor type not stated	–	none mentioned	undated project page; API repo at commit a0936196, fetched...	the page claims 21 DoF with 17 powered motors and the README calls the same hand 17-DOF
sanctuary_phoenix_hand_2024	Sanctuary AI (Sanctuary Cognitive Systems Corporation)	hydraulic, 'unique miniaturized hydraulic valves'	–	not in the press release text; sidebar-only headlines mention new touch/tactile sensors as a separate later announcement, not...	undated press release; trade reprint 2024-12-17	–
ilda_hand_2021	Ajou University, Korea Institute of Machinery & Materials, Korea University	linkage-driven, direct linear drive (3 Maxon DCX8M motors per finger with GPX8 16:1 gearboxes driving ball screws via parallel+serial four-bar...	fingertip normal force in the bent pose; 28 N stretched, and 25 N per finger measured while crushing an...	6-axis F/T sensor per fingertip (5 total), force resolution 62 mN, range +/-35 N; not a distributed taxel skin	paper, accepted 2021-11-05	–
ruka_v2_2026	Xinqi (Lucas) Liu, Ruoxi Hu, Alejandro Ojeda Olarte, Zhuoran Chen, Kenny Ma, Charles...	tendon-driven, forearm-mounted actuators; decoupled parallel 2-DoF wrist via a passive spherical ball joint; single-tendon-plus-spring...	–	optional e-flesh fingertips (magnetic touch-sensing form factor); not part of the base design	paper, 2026	the AS5600 encoders are attachable and detachable and are read for calibration and measurement only, not in...
tesollo_dg5f_2024	Tesollo Inc. (Incheon HQ, Gwangmyeong R&D/factory)	one integrated actuator per joint ('high-torque actuation', absolute encoder); page does not say direct-drive	–	none standard; optional fingertip sensors (6-axis F/T, 3-axis force, or tactile) available; count/type not given	undated page, copyright 2026, download links dated 2026-08	–
unitree_dex5_2025	Unitree Robotics (Yushu Technology Co., Ltd.)	in-joint geared motor ('hollow-cup motor' + high-precision encoder + low-damping small-clearance reducer), backdrivable; page does not use the words...	fingertip normal force under a 1 cm diameter cylinder pressed vertically down (vendor footnote)	Dex5-1: none. Dex5-1P: 94 pressure sensors per hand (2x5 palm + 2x3x5 finger pad + 2x3x5 fingertip + 2x3x4 finger root), range 10...	undated page, copyright 2016-2025, fetched 2026-09-17	–
wuji_hand_2025	Wuji Technology (founded 2019)	direct-drive rotary, backdrivable	–	'Multi-Axis Force/Torque Fingertip Sensing' confirmed present; no type, count or range given	undated page, copyright 2026, fetched 2026-09-17	the page footnotes that these figures 'reflect the current Beta1 product and the SPEC will continue to...

continued on the next page

Table XI, continued.

hand key	maker, in full	actuation, as the source describes it	what the force figure is	tactile, as the source describes it	date of the claim	caveat on the row
pisa_iit_softhand_2014	Centro E. Piaggio, University of Pisa and IIT (Catalano, Grioli, Farnioli, Serio, Piazza,...)	tendon-driven adaptive synergy, single motor via differential gears (6 W Maxon RE-max21, 84:1 reduction)	–	none (motor encoder only; a padded work glove supplied contact compliance during experiments)	paper, IJRR 2014	–
orca_hand_2025	Clemens C. Christoph, Maximilian Eberlein, Filippus Katsimalis, Arturo Roberti,...	tendon-driven (antagonistic fishing-line tendon pairs per joint); wrist uses a GT2 timing belt drive	–	yes: FSR-based binary tactile sensing on all 5 fingertips (RP-C7.6-ST), absolute detection threshold as low as 0.05 N	paper, 2025	the 19.6 N previously tabulated as a fingertip force is the newton equivalent of the 2 kg index-finger...
allegro_hand_v4_2016	Wonik Robotics Co. Ltd. (Seoul, South Korea); earlier versions 1.0/2.0 made by SimLab Co...	–	–	none	driver repo, versions 1.0-4.0 undated, fetched 2026-09-17	allegrohand.com/v4 returned HTTP 404 and the Wonik wiki timed out; every cell here comes from the ROS driver...
faive_hand_2023	Yasunori Toshimitsu, Benedek Forrai, Barnabas Gavin Cangan, Ulrich Steger, Manuel Knecht,...	tendon-driven, rolling-contact joints (16 Dynamixel XC330-T288-T servos, 6 antagonistic pairs)	–	none on the physical hand; a simulated 'fingertip force' (15-dim, critic-only/privileged) exists only in the RL training...	paper, 2023	the row's open-hardware status is unconfirmed: the repo has LICENSE and LICENSE-NVIDIA files whose contents...
leap_hand_2023	Kenneth Shaw, Ananye Agarwal, Deepak Pathak, Carnegie Mellon University	direct-drive (Dynamixel servos, e.g. XC330-M288), joint velocity ~8 rad/s	pull-out resistance: the outward force a flexed finger resists before slipping or deviating more than 15...	none (future work only: 'we plan to develop and integrate LEAP Hand with low-cost touch sensors')	paper, 2023	–
ruka_2025	Anyia Zorin, Irmak Guzey, Billy Yan, Aadithya Iyer, Lisa Kondrich, Nikhil X. Bhattasali,...	tendon-driven (11 Dynamixel actuators in the forearm: XM430-W210T for the thumb, XL330-M288-T for the other fingers)	pinch force, best of three trials, averaged over the left and right hands (Table III)	none (stated limitation: 'lacks tactile sensing')	paper, 2025	–
inspire_rh56dfx_2023	Beijing Inspire Robots Technology Co., Ltd.	–	fingertip strength, vendor spec, four fingers; thumb 15 N; force resolution 0.50 N	none (this variant is the 'without tactile sensors' table; a tactile FTP variant exists per bib but is not on this page)	undated page, fetched 2026-09-17	the vendor's own page reads 'Degrees of freedom 6, Numbers of joints 12'; the joints column here prints the...
robotera_xhand1_2024	ROBOTERA	gear-driven force-controlled joint modules per finger segment, back-drivable	–	tactile/force sensors on every fingertip confirmed present (senses contact, force, temperature); count and array size not stated	undated tracker page, fetched 2026-09-17; the vendor page...	the 15 N in circulation comes from a survey table, not from the tracker or the vendor; the tracker states no...
shadow_dex_ee_2024	Shadow Robot Company, – in collaboration with Google DeepMind	–	–	stereo camera-based fingertip tactile sensors (hundreds of taxels each); multi-taxel 3-DoF arrays on middle and proximal...	undated page, copyright 2026, fetched 2026-09-17	–
brainco_revo2_2025	BrainCo Inc.	–	pinch force, vendor minimum (≥ 15 N); the same page states a whole-fist grip of ≥ 50 N	Touch variant only: multi-dimensional fingertip tactile module with 'Tactile Adaptive Control'; count/type not stated. Basic and...	undated page, fetched 2026-09-17	–
agibot_omnihand_2025	–	–	–	–	no source: both store.agibot.com URLs 404, fetched...	–
boston_dynamics_atlas_hand_2026	Boston Dynamics	electric (robot is fully electric); specific hand mechanism not stated	–	tactile sensing in the fingers and palms, confirmed present; type and count not stated	trade press, 2026-01-05 (CES); the Boston Dynamics blog of...	–
daxo_muscle_v0_2025	Daxo Robotics	ultra-redundant tendon-driven; compliant structure with no rigid joints, flexible materials and tendon routing	–	–	third-party catalogue post, 2026-09-16; daxo-robotics.com...	120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin; no DoF...
dexhand_open_source_2023	Rob Knight, The Robot Studio (design); electronics/firmware/ROS 2 by Trent Shumay, IoT...	tendon (fishing line, Sufix 832 80lb / 0.8mm kitleline); Emax ES3301/ES3302/ES3351/ES3352 micro-servos for fingers, Feetech SCS2332 or PWM servos for...	–	none mentioned	page posts 2023-08-08 to 2023-10-01; no hardware release...	no joint count is given anywhere, so no DoF figure can be quoted; the servo count is 16 finger and thumb...
figure_03_hand_2025	Figure AI	–	–	not stated on page for the hand itself; palm cameras in each hand are vision, not tactile	undated tracker page, fetched 2026-09-17; figure.ai launch...	the tracker's 'Degrees of freedom, hands 20' sits beside 'Number of fingers 10', so the 20 is most likely...

continued on the next page

Table XI, continued.

hand key	maker, in full	actuation, as the source describes it	what the force figure is	tactile, as the source describes it	date of the claim	caveat on the row
linkerbot_ l20_2025	LinkerBot (SDK author 'CHIUS INC')	—	—	L10/L20: per finger 4 channels (normal force, tangential force, tangential direction, proximity), 5 sites x 4 channels, 0-255...	undated page, fetched 2026-09-17	—
paxini_ dexh13_2024	PaXini Tech	—	—	1,140 ITPU multidimensional tactile processing units (press-release claim; unclear if PX-6AX-GEN3 sensor family)	press, CES 2026-01-07	—
psyonic_ ability_ hand_2021	PSYONIC (San Diego, CA, USA)	linkage (repo file/function names indicate a four-bar finger linkage; not stated on the vendor page itself)	—	30 touch sensor values streamed (FSR-based per API and URDF 'no_fsr' variant); per-finger split not stated	undated page, fetched 2026-09-17	the four-bar finger linkage rests on repository file names and the five fingers on URDF meshes; the vendor...
tesla_ optimus_ hand_2025	Tesla	V3: tendon-driven from forearm actuators, three tendons per finger through a crosstalk-managing wrist. Gen 2: in-hand actuators/sensors (unlocated in...	—	Gen 2: tactile sensing on all fingers, demonstrated matching contact location and force in an egg-pickup video; type/count not...	Gen 2 video 2023-12-13; V3 patents filed 2024-10, relayed...	the 22 DoF in circulation is a Teslarati article's arithmetic of four DoF on each of five fingers plus two at...
xiaomi_ cyberone_ hand_2026	Xiaomi	motors located in the hand (compact motors generate heat requiring liquid cooling); no tendon/linkage statement, though contrast with tendon-driven...	—	full-palm tactile sensing, area ~8,200 sq mm, detects pressure and contact across the whole palm not just fingertips; taxel count...	press, 2026-03-30	—

TABLE XII: The contact-model and solver descriptions behind the controlled vocabulary of Table III, verbatim from the parsed source. These cells run to several hundred characters each, which is why the body table prints a vocabulary instead: a 480-character cell in a body float is a paragraph printed sideways. 4 of the 30 cells in the 2 description columns (13%) are $\rightarrow$: no source stated the value. Nothing here is cut.

engine	contact model, as the source describes it	solver, as the source describes it
brax_2021	rigid bodies in maximal coordinates; naive/quadratic-scaling pairwise collision detection (no broad-phase acceleration structure); no convex-decomposition or mesh-vs-primitive distinction described	not LCP-based; velocity-level collision updates with Baumgarte stabilization (inspired by the Tiny Differentiable Simulator); joints modeled as spring constraints rather than Featherstone-style articulated-body methods
comfree_sim_2026	complementarity-free, analytical (closed-form) contact resolution in the dual cone of the Coulomb friction cone via a prediction-correction/impedance-style update, extended to a unified 6D model (normal, tangential, torsional, rolling friction)	no complementarity solve; a 4-kernel GPU pipeline (Algorithm 1: smooth-velocity prediction, per-contact/per-face dual-cone solve, generalized-impulse accumulation, velocity correction) computes the impulse in closed form per contact facet
dojo_2022	hard-contact nonlinear complementarity problem (NCP) with an exact nonlinear (second-order-cone) friction cone, avoiding pyramidal/linearized approximation, to avoid interpenetration and creep artifacts	custom primal-dual interior-point solver (Algorithm 2), based on Mehrotra’s predictor-corrector algorithm, extended for non-Euclidean quaternion variables, with cone-handling borrowed from CVXOPT; gradients via implicit differentiation (implicit function theorem) of the KKT/complementarity system, with central-path parameter kappa trading gradient smoothness for physical accuracy
genesis_2024	Constraint-based rigid solver (equality/inequality constraints, plus an explicit noslip() post-pass, architecturally similar to MuJoCo’s solver+noslip design) combined with multi-material coupling: FEM, MPM, particle-based PBD and SPH, a libuipc Incremental Potential Contact solver, an explicit inter-solver coupler, and a SAP (semi-analytic-primal-style) compliant-contact path used for grasp-coupling scenes.	ConstraintSolver with Newton-style line-search iterative resolve (linesearch.py) and constraint-island decomposition for parallelism (island.py); multiple pluggable physics solvers (Rigid, FEM, MPM, PBD, SPH, libuipc, SAP) compiled via the Quadrants (Taichi-derived) backend.
isaacgym_2021	Rigid-body contacts via PhysX; contact geometry can be primitive shapes or meshes loaded from URDF/MJCF (Sec. 2.2); net contact forces exposed per-rigid-body via the Tensor API (Table 1, Get-only). No explicit statement of convex-vs-mesh contact generation or per-pair contact-point count. Shadow Hand tendons simulated via PhysX Fixed Tendon mechanics: spring+damping force proportional to deviation from rest length, propagated through a tendon-joint tree (Sec. 3, Appendix A.1).	Temporal Gauss-Seidel (TGS) solver (Sec. 3, ref. 18), not classic PGS/LCP: folds sub-stepping into a per-body accumulated velocity-delta buffer projected onto the constraint Jacobians. Table 3 exposes two user-tunable iteration knobs, Position iterations (biased, velocity+positional-error-correcting) and Velocity iterations (unbiased, velocity-error-only-correcting); no default numeric values for these are given in the main text.
isaacsim_2025	Rigid contacts by default via PhysX 5 (SDF collisions, Featherstone articulation solver, Sec. 2.2); ‘Factory’ assembly envs specifically use ‘SDF-based contact generation, a contact reduction technique, and a Gauss-Seidel solver’ (Sec. 6.4). ContactSensor exposes net normal force per body, optional filtered pairs, contact duration, average contact point, and a short history (Sec. 3.3.1); no contact-quality/penetration metric is defined (survey note).	NVIDIA PhysX 5’s internal rigid-body/Featherstone-articulation solver, used as-is; Factory assembly envs specifically named as using a Gauss-Seidel solver with SDF-based contact generation and contact reduction (Sec. 6.4). No timestep, decimation, or solver-iteration values are stated anywhere in the paper for the general (dexterous) suite; the only note is qualitative, that the Digit humanoid ‘requires a higher solver iteration count for stable simulation’ (Fig. 14).
maniskill3_2024	PhysX (via SAPIEN) rigid-body contacts on GPU; no statement of convex-vs-mesh contact generation or contact-point count in the note. AllegroHandRightTouch reads FSR contact impulse (‘get_fsr_impulse’) for touch sensing (App. VII-H, code).	PhysX (via SAPIEN), used as-is; no LCP/PGS/TGS/Newton solver-algorithm name given. Benchmark (cartpole) settings state ‘Solver Position Iterations: 4 / Solver Velocity Iterations: 0’ (App. XI-A), but per-task ‘_default_sim_config’ values are not exported in the parsed source.
mujoco_2012	soft, convex velocity-stepping contact; three interchangeable solvers replace the standard LCP-with-friction-pyramid approach: implicit-complementarity, convex (kinetic-energy minimization with a soft non-penetration cost), and diagonal (mass-aware spring-damper)	implicit-complementarity solver (customized non-smooth Newton method, most accurate); convex solver (interior-point method used for reported timings; projected Newton/CG/Gauss-Seidel also mentioned as unverified-faster alternatives); diagonal solver (per-contact critically-damped spring-damper using the diagonal of the A matrix, fastest/least accurate)
mujoco_convex_contact_2014	soft, convex, complementarity-free; a unified impulse vector covers joint dry friction, joint/tendon/distance limits, and frictional contacts together, with an elliptical friction cone whose dimensionality per contact is 2 (sliding), 3, or 5 (sliding+torsional+rolling)	GPGS (generalized projected Gauss-Seidel, described as the authors’ own unpublished-at-the-time method handling cone and pyramid constraints) for the contact/limit impulse; a separate ‘soft Gauss principle’ derivation for equality/holonomic constraints; an analytically-invertible closed-form solver for a restricted circular-cone special case
mujoco_warp_2025	MuJoCo’s native solref/solimp-parameterized soft-constraint contact model ported to Warp — per-contact Jacobian construction (dense and sparse variants, with flex variants) and equality/limit/friction constraint rows; not a re-derived contact model	MuJoCo’s default (implicit CCP-style) solver and Newton constraint solver (via a ‘newton’ flag threaded through equality/limit/friction kernels) are supported; legacy PGS solver and noslip post-processing pass are ‘not yet supported’ per the README
newton_2025	Solver-dependent: MuJoCo-style soft-constraint (solref/solimp-style) contact via SolverMuJoCo/MJWarp; Proximal-ADMM / Dual-Variational-Inequality contact resolution via SolverKamino; position-based-dynamics-style constraint projection for the XPBD and VBD solvers.	Multi-solver architecture: SolverMuJoCo (wraps MJWarp), SolverKamino (Proximal-ADMM + DVI), XPBD (rigid), VBD (cloth/deformables), Style3D (cloth), plus a separate IKSolver (LM and L-BFGS backends).
orbit_2023	PhysX SDK 5 signed-distance-field (SDF) collision checking for rigid bodies (handles non-convex geometry such as screw threads); FEM-based solver using a constraint-based stable neo-hookean material formulation for deformable bodies; PBD (position-based dynamics) solver for cloth	PhysX SDK 5’s internal rigid-body and FEM solvers, used as-is; no LCP/PGS/TGS, Newton, or CG iteration counts given
pybullet_2016	—	—
raisim_2018	—	—
sapien_2020	Rigid-body contact with convex-decomposed collision meshes (PhysX 4.1); three joint systems offered (kinematic, dynamic, and PhysX articulation) trading control accuracy against speed.	PhysX 4.1’s internal rigid-body solver, used as-is; no LCP/PGS/TGS name or iteration count given in the paper.

TABLE XIII: The two columns of the protocol that Table VII leaves out, verbatim from the generator that derives every count in it. No cell is –. Read with Table VII: the axis names match row for row.

axis	how the measurement is made	why the count is what it is
Task success	criterion written before the run by the designer and scored by someone else; terminal predicate held for a 0.5 s dwell; initial conditions matched by image overlay; policies interleaved blind in one session	a rate without a denominator and an interval cannot be compared; the rubric separates a near-miss from inaction; the design is paired, so the count that governs it is paired
Robustness	axes that must not change the action and axes that must be scored separately; ranges stated; anchor replayed exactly	the ranking at the anchor is not the ranking under shift; at 40 per arm a fall from 0.50 to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1
Unseen objects	objects drawn from a stated distribution disjoint from training; bootstrap over objects, not trials	the resampling unit is the object, so trial counts overstate the precision
Physical plausibility	dense surface sample against the object mesh or SDF, computed on the policy's own rollouts by code that never entered the reward or the termination rule	a measure the policy optimised is not evidence about the policy
Sample and wall-clock cost	counted to that checkpoint, not to the end of training; GPU model and count stated	training variance is not rollout variance and the two are routinely conflated
Real-robot transfer	the same 100 initial conditions run in both; correlation on the pairs; variance ratio bootstrapped over the same pairs	a simulator earns a real-world claim only through its measured paired correlation; at 100 pairs $\hat{\rho} = 0.70$ carries 0.58 to 0.79, enough to say a simulator tracks reality at all and not enough to separate <code>suressim_2025</code> 's 0.70 regime from its 0.59 one
Reproducibility	diff the paper's stated objective against the released config and name the file and line of every disagreement	16 of the 62 code-releasing method rows carry a true contradiction between the paper's stated objective and the released code, and 38 carry a recorded disagreement of any kind (§5.8)

[illegible]